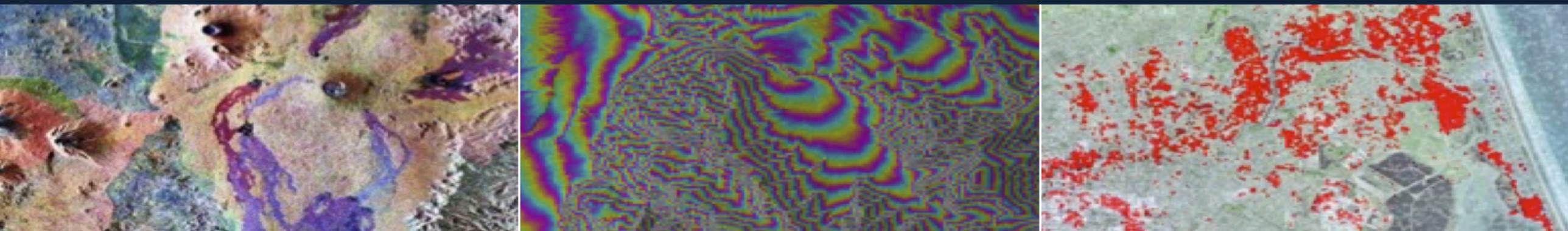
SYNTHETIC APERTURE RADAR INTERFEROMETRY · LECTURE 2

Geometric Aspects of InSAR

AKSHAY RAJ MANOCHA



THIS LECTURE

Key concepts

01 How the phase of a radar wave relates to the distance it travels

02 Wave interference: how the relative phase of two waves meeting at a point forms a pattern

03 How the relative phase of two observations from different vantage points relates to topography

04 How the relative phase of two observations at different times relates to surface displacement

05 Why interferometry is far more sensitive to displacement than to topography

BEFORE WE START

Review the SAR lecture

This lecture uses equations, variables and concepts from the SAR lecture. Can you explain each of these?

- $s(\rho, x)$
- Backscatter
- Range
- Azimuth
- Layover
- Shadow
- Speckle
- Resolution

If not, review Lecture 1 before going further.

01

PART 1

Interferometry

Phase and distance · wave interference · SAR interferometry · the interferogram

INTERFEROMETRY

Phase is proportional to distance

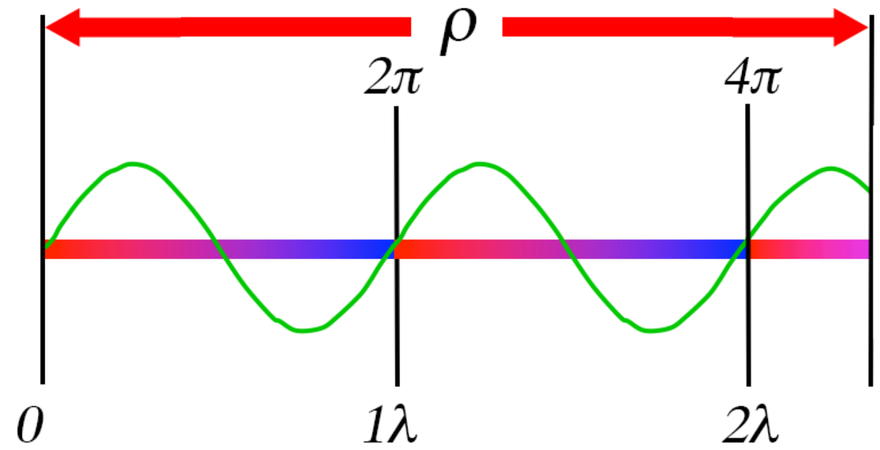
$$\phi = 2\pi \cdot \frac{\rho}{\lambda}$$

2π
radians per wavelength

ρ / λ
number of wavelengths

ρ: distance along the propagation path · λ: wavelength · φ: phase of the wave

The absolute phase at a point is hard to determine, but the **relative phase**, the difference in phase between two waves measured at a point, is much easier to measure.



RADAR INTERFEROMETRY

The use of phase measurements to find the relative distance to an object imaged by SAR from two or more observations, separated in time or in space.

Interferometric phase is simply another way of measuring relative distance. Phase can be measured to about a degree; with radar wavelengths of 3–80 cm, that means relative range accuracy at the millimetre level.

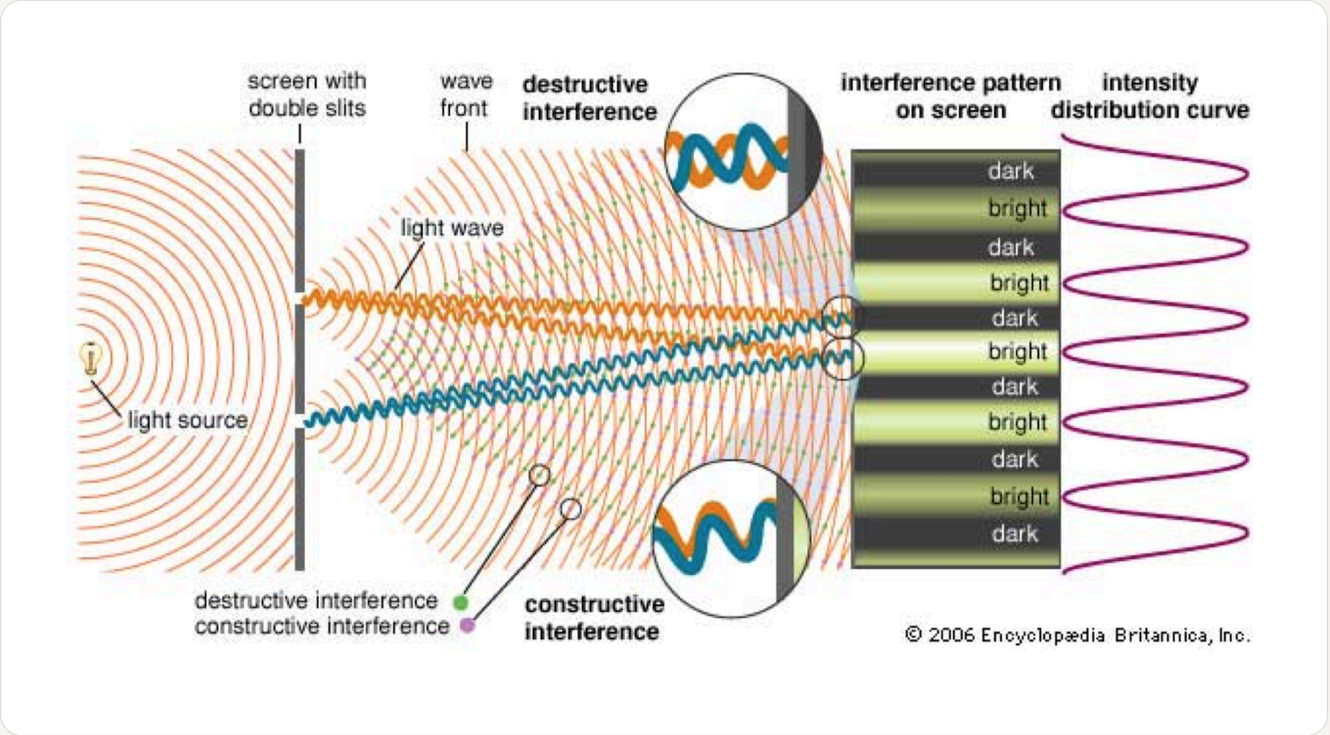
INTERFEROMETRY

Wave interference

The phase difference of two waves at a point is proportional to the difference in the distances they travelled to reach it.

Along a surface, these phase differences form an **interference pattern** that records the different paths the waves took.

The wave amplitude varies sinusoidally in time, but the relative phase at each point is constant: a fixed value between −π and π.



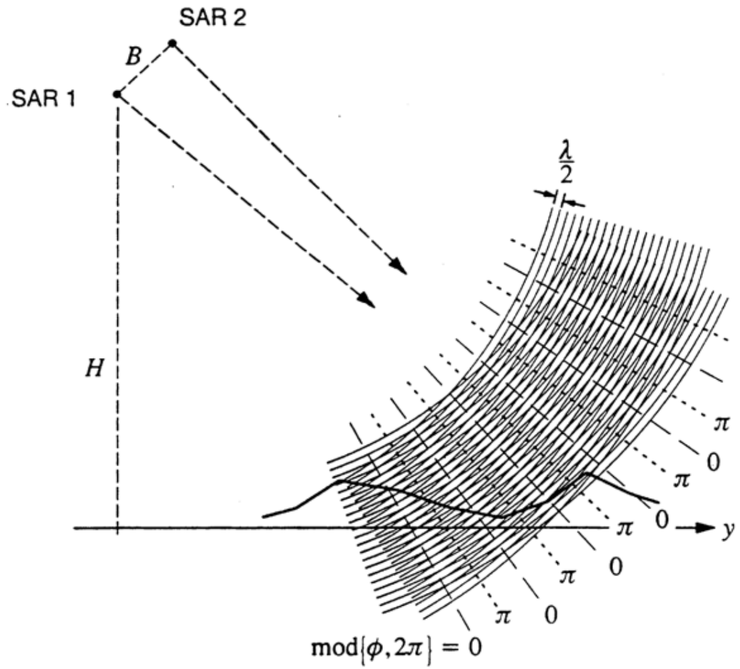
INTERFEROMETRY

SAR interferometry

A SAR image is made by transmitting a wave and receiving its echo from the ground as the radar flies along its path.

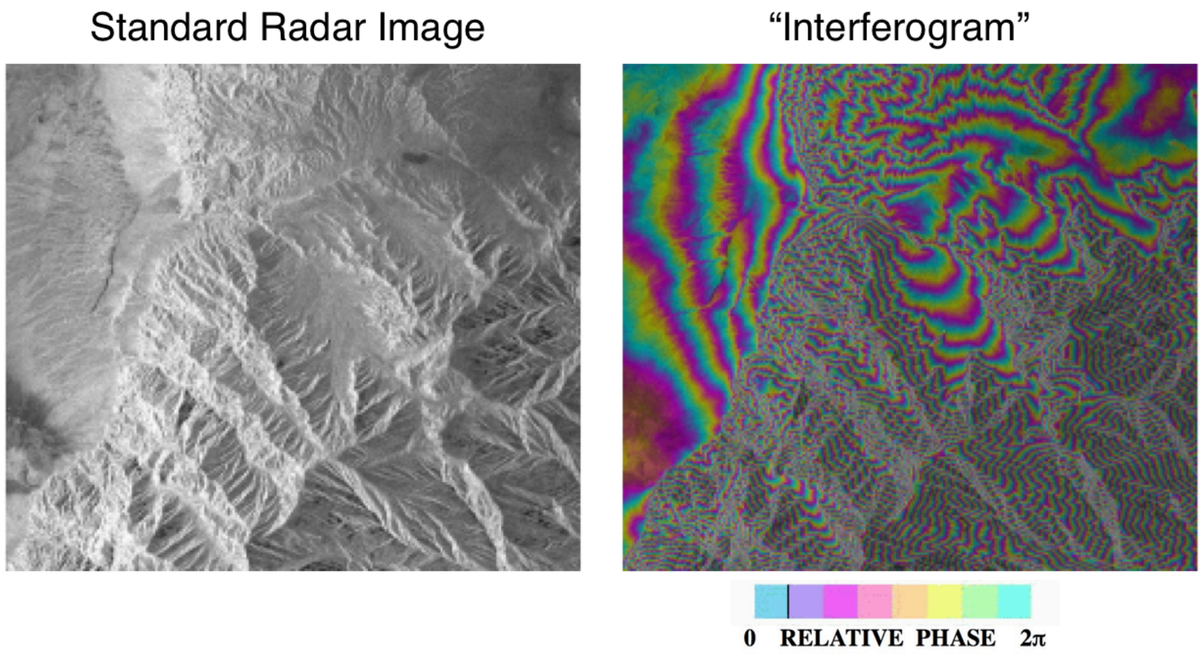
The phase of the wave at each point on the surface depends on the distance from the radar to that point.

A second radar at a different vantage point sees slightly different distances. The **relative phase** between the two observations encodes the relative distance, which varies with the topographic relief.



INTERFEROMETRY

The SAR interferogram



Left: a standard SAR image (radar reflectivity). Its phase alone would look random because of speckle.

$$i(\rho, x) = s_1(\rho, x)\, s_2^*(\rho, x)$$
$$= |s_1|\,|s_2|\, e^{\,j(\phi_1 - \phi_2)}$$

The interferogram is the complex product of one image with the conjugate of a second, taken from a slightly different vantage point.

ρ: range · x: position along the flight direction · s_1, s_2: the two complex images

One colour cycle = 2π of relative phase. Magnitude relates to correlation; phase to differences in observation geometry.

02

PART 2

Interferometry for topography

The interferometric triangle · baselines · fringe rates · flattened phase · ambiguity height

TOPOGRAPHY

The interferometric triangle

The measured phase difference in the interferogram is

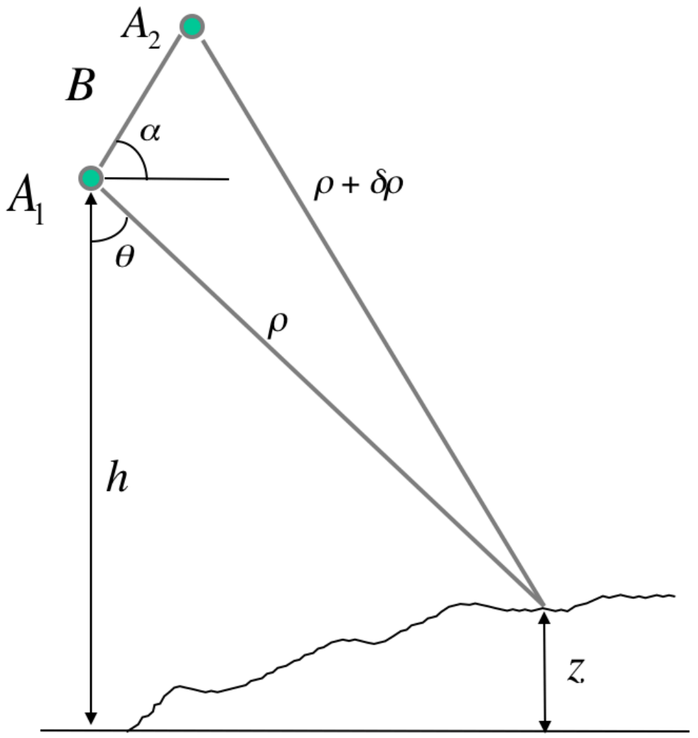
$$\Delta\phi = \phi_1 - \phi_2 = \frac{2\pi p}{\lambda}\,\delta\rho$$

where p = 1 or 2, depending on how the data were collected. Having measured Δφ, solve for δρ, then triangulate with the law of sines:

$$\sin(\theta - \alpha) = \frac{(\rho + \delta\rho)^2 - \rho^2 - B^2}{2\rho B}$$

$$z = h - \rho\cos\theta$$

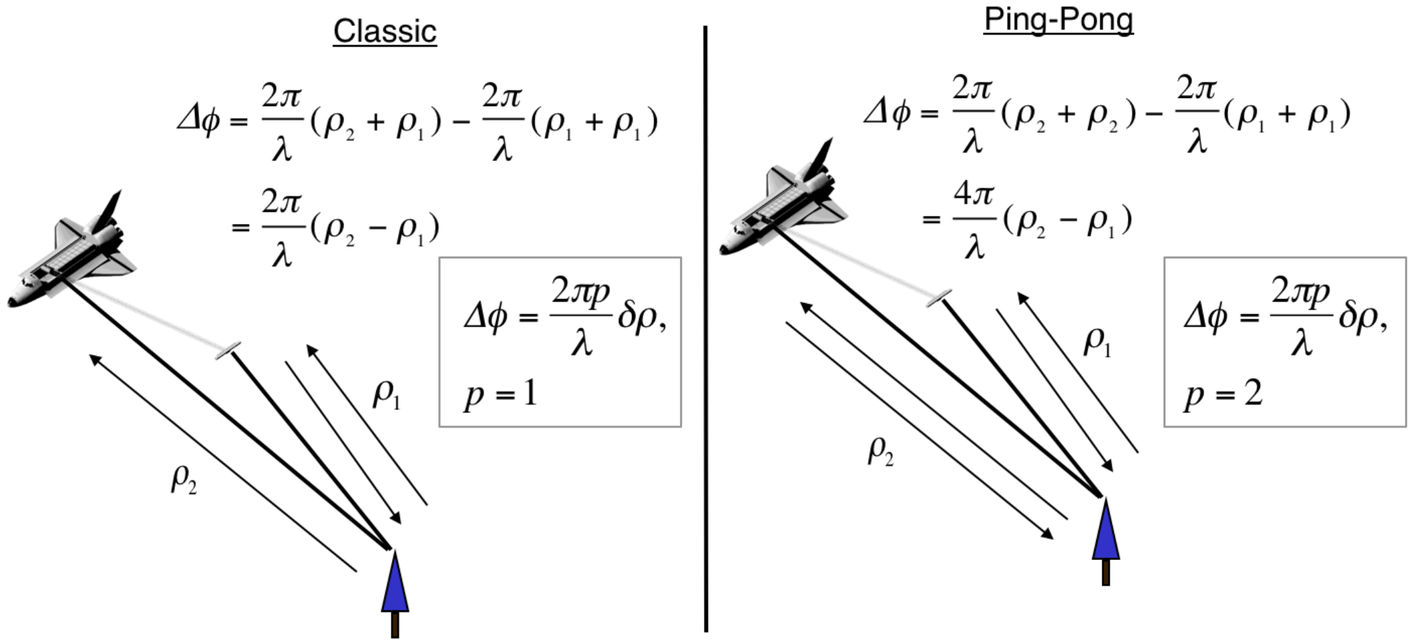
> Accuracy needed: B known to mm, α to the equivalent angle, ρ to metres, and the system phase-stable to mm.



TOPOGRAPHY

How the data are collected sets p

Single-pass interferometry: two antennas observe at the same time, on one platform or on two co-flying platforms.



Classic · p = 1

One antenna transmits and both antennas receive.

Ping-pong · p = 2

Each antenna transmits and receives its own echo, effectively doubling the physical baseline.

TEST YOUR UNDERSTANDING

Interferometer methods

What is the value of p for an interferometer formed by a single radar that repeats its orbit track closely, comparing the first observation with the second (**repeat-pass** interferometry)?

A p = 3

B p = 0

C p = 1

D p = 2

ANSWER

D: p = 2

On each pass the radar transmits and receives its own echo, so each path is a full round trip from that pass's position, just as in ping-pong mode. The path difference counts twice, so p = 2.

$$p = 2$$

repeat-pass, like ping-pong

Hint from the notebook: assume the surface is static over time.

TOPOGRAPHY

Phase and the interferometric baseline

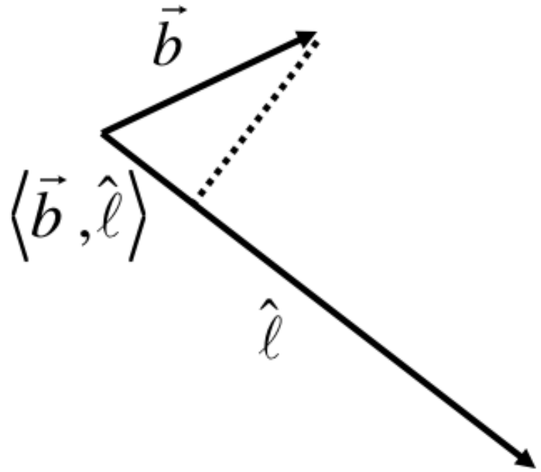
The interferometric phase can be approximated as

$$\Delta\phi \approx -\frac{2\pi p}{\lambda}\left\langle \hat{\ell}, \vec{b} \right\rangle$$

$$\vec{b} = (b\cos\alpha,\ \ b\sin\alpha), \qquad \hat{\ell} = (\sin\theta,\ \ -\cos\theta)$$

$$\Delta\phi \approx -\frac{2\pi p}{\lambda} b\sin(\theta - \alpha)$$

$\langle\ ,\ \rangle$ is the dot product. The phase is proportional to the projection of the baseline vector b onto the unit look vector $\hat{\ell}$, often called the **parallel component of the baseline**.



TOPOGRAPHY

The phase gradient

For flight planning and phase-unwrapping studies, we want the interferometric phase gradient for any surface slope, look geometry and baseline. The phase is well approximated by the first line; its gradient follows:

$$\Delta\phi = -\frac{2\pi p}{\lambda}\left\langle \vec{b}, \hat{\ell} \right\rangle$$

$$\frac{\partial \Delta\phi}{\partial s} = -\frac{2\pi p}{\lambda}\left\langle \vec{b}, \frac{\partial \hat{\ell}}{\partial s} \right\rangle \qquad \frac{\partial \Delta\phi}{\partial \rho} = -\frac{2\pi p}{\lambda}\left\langle \vec{b}, \frac{\partial \hat{\ell}}{\partial \rho} \right\rangle$$

b: baseline vector · ℓ̂: unit vector pointing to the target · λ: wavelength · ⟨ , ⟩: dot product · s: along-track coordinate · ρ: range

TOPOGRAPHY

Fringe rate across and along track

Across track: the phase changes with range even on flat terrain. This flat-surface (spherical-Earth) fringe frequency depends on the local slope and the perpendicular baseline b⊥.

$$\frac{\partial \Delta\phi}{\partial \rho} = \frac{2\pi p}{\lambda} \frac{b\cos(\theta - \alpha)}{\rho \tan(\theta - \psi_c)} = \frac{2\pi p}{\lambda} \frac{b_\perp}{\rho \tan(\theta - \psi_c)}$$

Along track: the fringe rate is zero unless there is an azimuth slope ψ_s; it grows with that slope and with b⊥.

$$\frac{\partial \Delta\phi}{\partial s} = \frac{2\pi p}{\lambda} \frac{b\cos(\theta - \alpha)}{\rho} \sin\theta \tan\psi_s = \frac{2\pi p}{\lambda} \frac{b_\perp}{\rho} \sin\theta \tan\psi_s$$

TOPOGRAPHY

Flattened phase

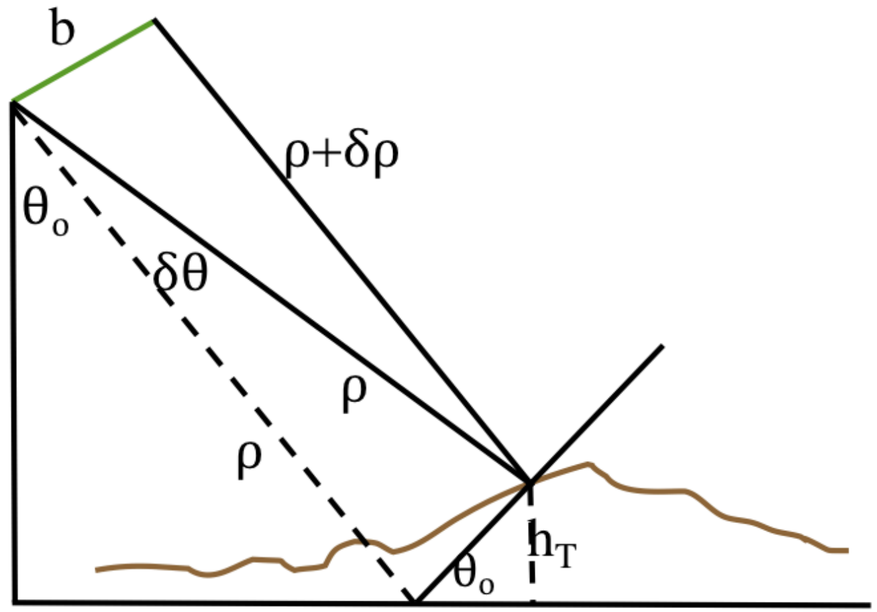
Many workflows remove the flat-Earth fringes, through motion compensation to a common reference or later in processing. What remains are fringes proportional to topography: the **flattened phase**.

$$\begin{aligned}\Delta\phi_{\text{flat}} &= -\frac{2\pi p}{\lambda}\left(\left\langle\hat{\ell},\vec{b}\right\rangle - \left\langle\hat{\ell}_0,\vec{b}\right\rangle\right)\\ &= -\frac{2\pi p}{\lambda}\, b\,(\sin(\theta-\alpha)-\sin(\theta_0-\alpha))\\ &\approx -\frac{2\pi p}{\lambda}\, b\cos(\theta-\alpha)\,\delta\theta \approx -\frac{2\pi p}{\lambda}\, b_\perp\,\delta\theta\end{aligned}$$

$\hat{\ell}$: look vector to a point · $\hat{\ell}_0$: look vector to the flat surface at the same range. The last two steps use the usual 2-D simplification.

TOPOGRAPHY

Flattened phase maps topography



From the geometry in the figure:

$$\delta\theta \approx \frac{h_T}{\rho \sin\theta_0}$$

$$\Delta\phi_{\text{flat}}(\rho, x) \approx -\frac{2\pi p}{\lambda} b_\perp \frac{h_T(\rho, x)}{\rho \sin\theta_0}$$

x is the position along the flight direction. The flattened phase at each pixel is proportional to the height h_T there.

TEST YOUR UNDERSTANDING

Ambiguity height

The **ambiguity height** is the change in elevation that changes the interferometric phase by 2π radians. Which formula defines it?

A $h_{\text{amb}} = \dfrac{p\, b_\perp}{\lambda \rho \sin\theta}$

B $h_{\text{amb}} = \dfrac{\lambda \rho \sin\theta}{p\, b_\perp}$

C $h_{\text{amb}} = \dfrac{\lambda\, p\, b_\perp}{\rho \sin\theta}$

D $h_{\text{amb}} = \dfrac{2\pi \lambda \rho \sin\theta}{p\, b_\perp}$

ANSWER

B: the ambiguity height

$$h_{\text{amb}} = \frac{\lambda \rho \sin\theta}{p\, b_{\perp}}$$

Set the flattened phase equal to 2π and solve for the height. Check the units: (m · m) / m = m.

693 m

per fringe, with λ = 0.24 m, ρ = 1,000 km, θ = 30°, b = 100 m, α = 0°, p = 2 ($b_{\perp}$ = 86.6 m)

03

PART 3

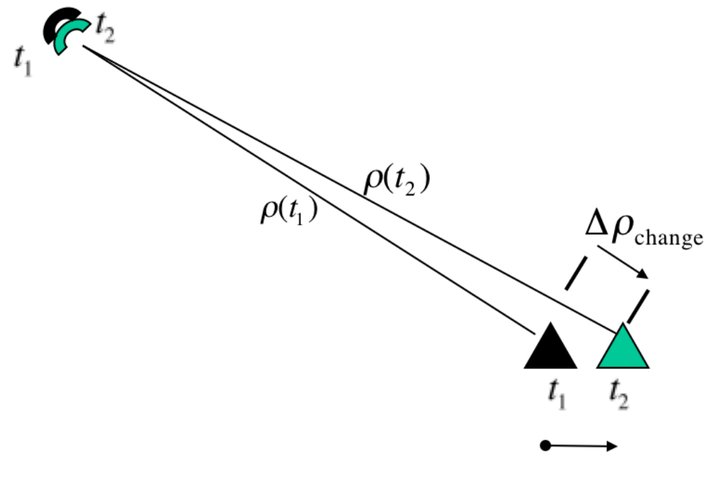
Differential interferometry

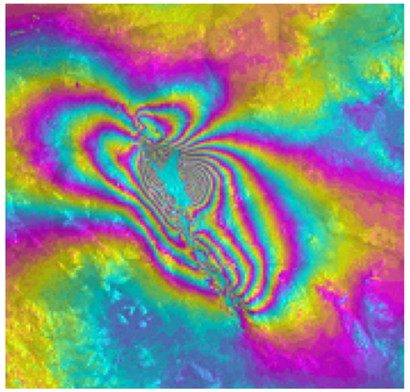
Repeat-pass interferometry · sign convention · topography vs displacement · sensitivity

DIFFERENTIAL INTERFEROMETRY

Repeat-pass interferometry

When two observations are made from the same position but at different times, the phase measures any change in range to the surface: its displacement along the line of sight.





$$\Delta\phi = -\frac{4\pi}{\lambda}(\rho(t_1) - \rho(t_2))$$
$$= \frac{4\pi}{\lambda}\,\Delta\rho_{\text{change}}$$
$$\approx \frac{4\pi}{\lambda}\left\langle \vec{D}, \hat{\ell} \right\rangle$$

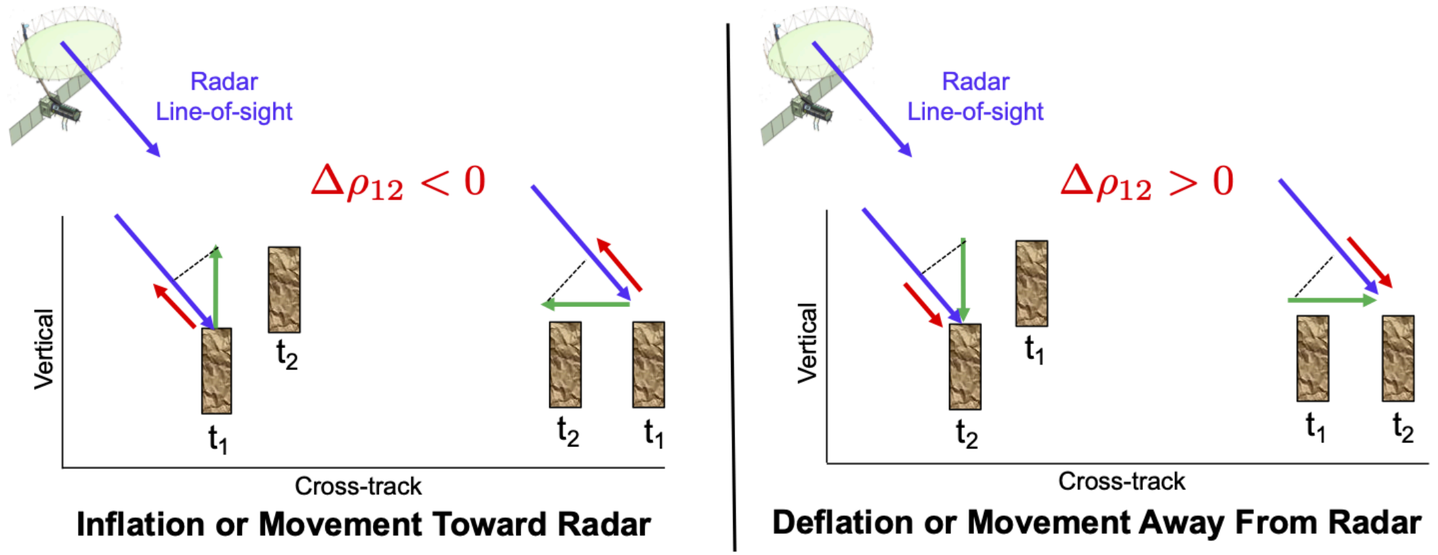
D: the vector displacement of the surface between the two times. The range change is its projection on the look direction $\hat{\ell}$.

DIFFERENTIAL INTERFEROMETRY

Deformation phase convention

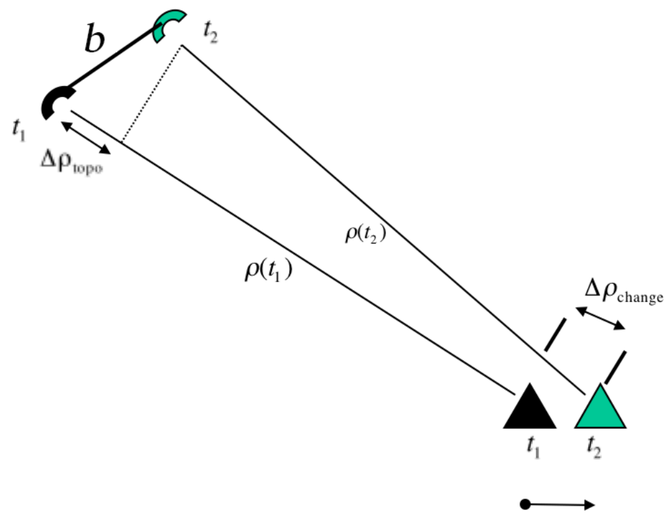
The usual convention conjugates the more recent image, so $t_2 > t_1$. With $\varphi = -4\pi\rho/\lambda$ for each image:

$$\Delta\phi = \arg(s(t_1)\, s^*(t_2)) = -\frac{4\pi}{\lambda}(\rho(t_1) - \rho(t_2)) = \frac{4\pi}{\lambda}(\rho(t_2) - \rho(t_1))$$



DIFFERENTIAL INTERFEROMETRY

Displacement and topography together



It is hard to repeat the observation position exactly, so the phase also has a topography component:

$$\Delta\phi = \frac{4\pi}{\lambda}\left[-\langle\vec{b},\hat{\ell}\rangle + \langle\vec{D},\hat{\ell}\rangle\right]$$

$-\langle\vec{b},\hat{\ell}\rangle$

Topography term

$\langle\vec{D},\hat{\ell}\rangle$

Displacement term

After flattening:

$$\Delta\phi_{\text{flat}}(\rho,x) = \frac{4\pi}{\lambda}\,\Delta\rho_{\text{change}}(\rho,x) - \frac{2\pi p}{\lambda}\,b_{\perp}\,\frac{h_T(\rho,x)}{\rho\sin\theta_0}$$

> A cm-scale range change can look like a metre-scale topography change. If topography is known, its term can be removed to reveal the displacement.

TEST YOUR UNDERSTANDING

Line-of-sight displacement

A single interferogram measures only the displacement along the line of sight: $\Delta\rho = \langle \vec{D}, \hat{\ell} \rangle$

For a polar-orbiting satellite looking right, perpendicular to its track, at a 30° incidence angle, which displacement components is an interferogram most sensitive to?

A East (i.e. right) and up

B North (i.e. into the page)

C East

D North and east

ANSWER

A: east and up

The look vector lies in the plane perpendicular to the north–south track. At 30° incidence, its components give the sensitivity to each direction of motion:

0.87	0.50	≈ 0
Up: cos 30°	East (or west): sin 30°	North–south: along the track

Hint from the notebook: radars point closer to nadir than to the horizon, so vertical motion dominates.

DIFFERENTIAL INTERFEROMETRY

Interferometric sensitivities

Topographic sensitivity

$$\frac{\partial \Delta\phi}{\partial h} = \frac{2\pi p\, b \cos(\theta - \alpha)}{\lambda\, \rho \sin\theta}$$

0.0091 rad/m

per metre of elevation change

Displacement sensitivity

$$\frac{\partial \Delta\phi}{\partial \Delta\rho} = \frac{4\pi}{\lambda}$$

52.36 rad/m

per metre of line-of-sight displacement

Since b/ρ ≪ 1, it takes a much larger change in h than in Δρ to change Δφ: here about **5,800 times** larger. Differential interferometry can detect millimetre-level deformation: “metre-scale topography, millimetre-scale topographic change”.

Notebook values: p = 2, b = 100 m, α = 0°, θ = 30°, ρ = 1,000 km, λ = 0.24 m

TEST YOUR UNDERSTANDING

Sensitivity

What happens to the sensitivities when the baseline goes to zero?

A Nothing changes

B Displacement sensitivity greatly reduces

C Topographic sensitivity vanishes

D Topographic sensitivity vanishes and displacement sensitivity greatly increases

ANSWER

C: topographic sensitivity vanishes

$$\frac{\partial \Delta\phi}{\partial h} = \frac{2\pi p\, b \cos(\theta - \alpha)}{\lambda\, \rho \sin\theta}$$

Proportional to b, so 0 when b = 0

With zero baseline both views are identical, so topography adds no phase.

$$\frac{\partial \Delta\phi}{\partial \Delta\rho} = \frac{4\pi}{\lambda}$$

Unchanged: 4π / λ

Displacement sensitivity does not depend on the baseline at all.

Hint from the notebook: set b = 0 in the code cell and recalculate.

04

PART 4

Phase unwrapping and error sources

Wrapped phase · unwrapping · the error budget · DEM, troposphere and baseline errors

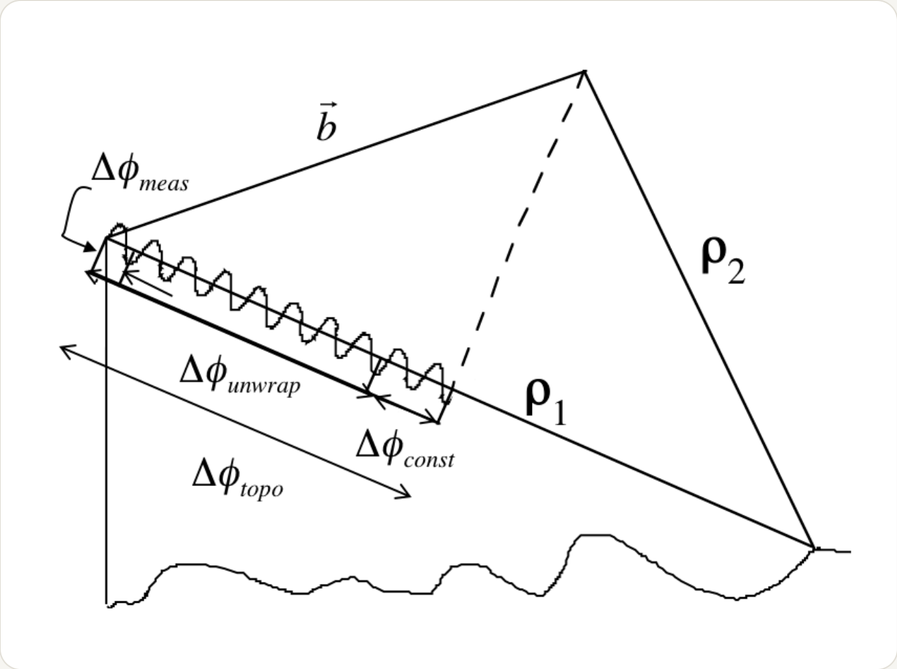
PHASE UNWRAPPING

Phase is measured only modulo 2π

The phase of a general interferogram is given by the first line, but it can't be measured directly. The interferogram i = |i| exp(jΔφ) is a complex image, so each pixel's phase is known only modulo 2π:

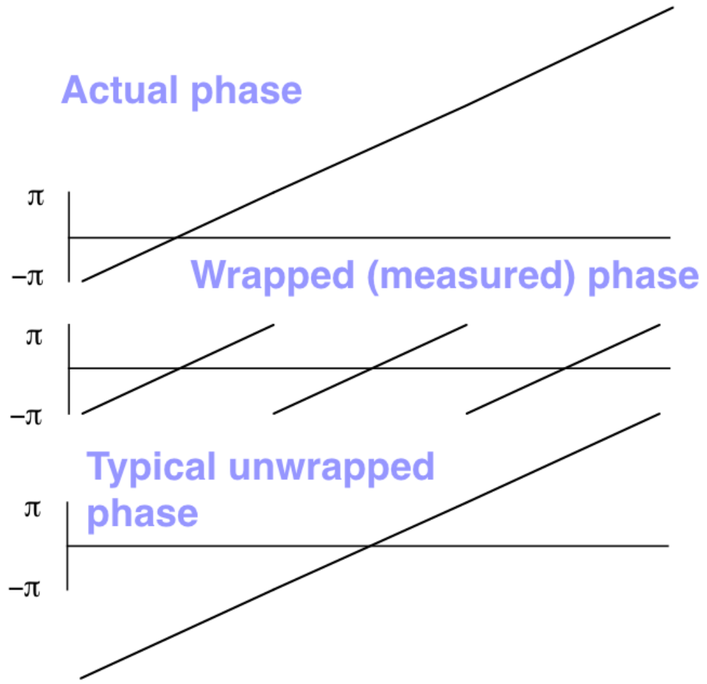
$$\Delta\phi = \frac{4\pi}{\lambda}\left[-\left\langle \vec{b}, \hat{\ell} \right\rangle + \left\langle \vec{D}, \hat{\ell} \right\rangle\right]$$

$$\Delta\phi_{\text{measured}} = \tan^{-1}\frac{\text{Im}(i)}{\text{Re}(i)} = \text{mod}(\Delta\phi,\ 2\pi)$$



PHASE UNWRAPPING

Unwrapping reconstructs the phase



Unwrapping reconstructs the phase that represents the physical range differences: topographic parallax, range change over time, or both.

$$\Delta\phi_{\text{unwrapped}}(\rho, x) = \Delta\phi(\rho, x) + \text{constant}$$

The one-dimensional example on the left is simple, yet even here the solution is known only to within a constant.

> In two-dimensional, noisy data, phase unwrapping can be quite challenging.

ERROR SOURCES

Interferometric error sources

Intrinsic to the technique

- Propagation delays from the atmosphere: troposphere and ionosphere
- Phase unwrapping errors
- Interferometric decorrelation

We will explore each of these in more detail in the course.

From the imaging-system design

Often controlled by the choice of look angles, antenna size, resolution and calibration:

- Layover and shadow from slant-range geometry
- Multiple scattering within and among resolution cells
- Range and azimuth ambiguities
- Range and azimuth sidelobes
- Multipath and channel cross-talk
- Calibration errors
- DEM errors

ERROR SOURCES

Residual errors after removing topography

A displacement interferogram whose topography was removed with a DEM still holds these errors on top of the signal:

$$\delta\Delta\rho = \frac{\lambda}{4\pi}\left[\Delta\phi_{\text{int}} + \frac{4\pi\, b_\perp\, \delta h_{\text{DEM}}}{\lambda\rho\sin\theta} + \Delta\phi_{\text{atm}}\right] + \left\langle \delta\vec{b},\, \hat{\ell}\right\rangle$$

$\Delta\phi_{\text{int}}$

Phase noise

Pixel-to-pixel noise: thermal noise relative to the signal, ambiguities, other multiplicative noise, interference.

$\frac{4\pi\, b_\perp\, \delta h_{\text{DEM}}}{\lambda\rho\sin\theta}$

DEM error

Height errors δh_DEM in the DEM, scaled by the perpendicular baseline.

$\Delta\phi_{\text{atm}}$

Atmosphere error

Path delays through the troposphere and the ionosphere.

$\left\langle \delta\vec{b},\, \hat{\ell}\right\rangle$

Baseline calibration

Error δb in the baseline; for spaceborne sensors usually a smooth linear or quadratic ramp in range and/or azimuth.

TEST YOUR UNDERSTANDING

Baseline error

Orbits are generally known to a few cm, varying slowly over the orbit. What would the phase residual be for a **3 cm baseline error** if the radar wavelength is **3 cm**?

A About 4π radians, but constant, so it does not affect relative displacement measurements

B A function of the topography, varying at the level of 4π radians

C About 4π radians, but varying smoothly across range

D Negligible, because 3 cm is much smaller than typical topographic variation

ANSWER

C: about 4π, varying smoothly across range

Size

(4π / λ) × 3 cm = 4π radians when λ = 3 cm.

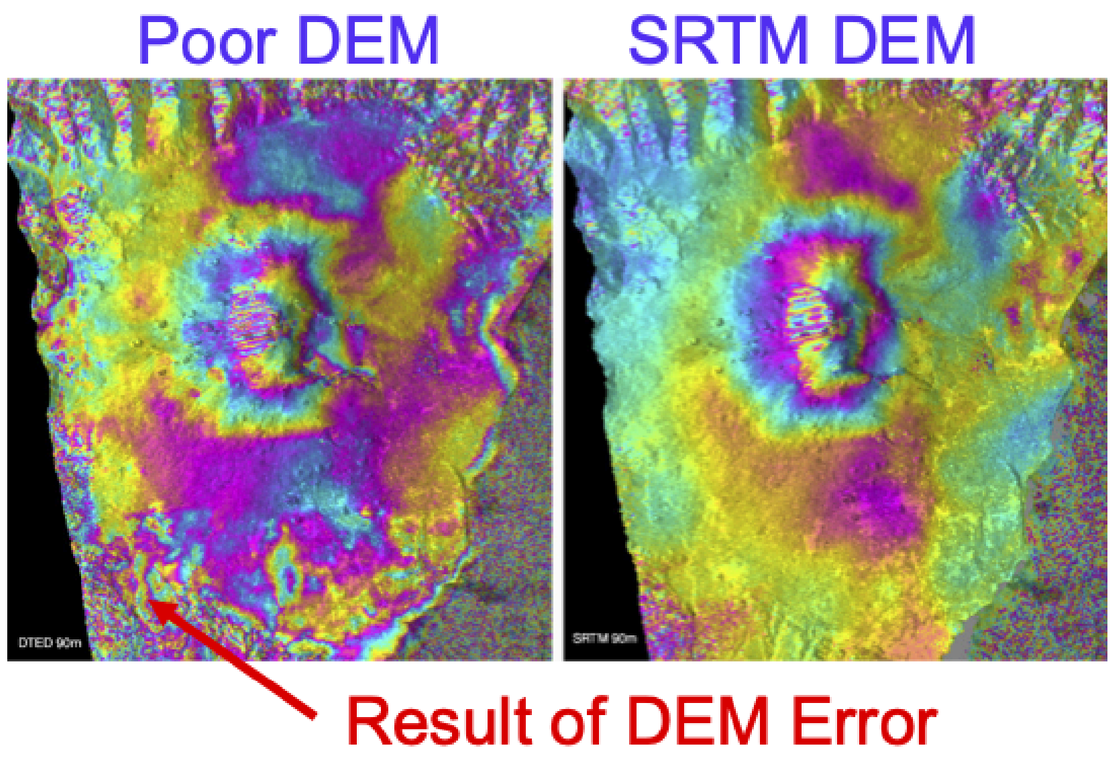
Shape

The look vector $\hat{\ell}$ changes across the swath, so $\langle \delta b, \hat{\ell} \rangle$ changes smoothly with range: a ramp, not a constant.

Hint from the notebook: is the line-of-sight vector constant across the swath?

ERROR SOURCES

DEM error example: Mount Etna



Phase errors appear when the DEM has height errors and the perpendicular baseline is non-zero, which is generally true somewhere in the interferogram.

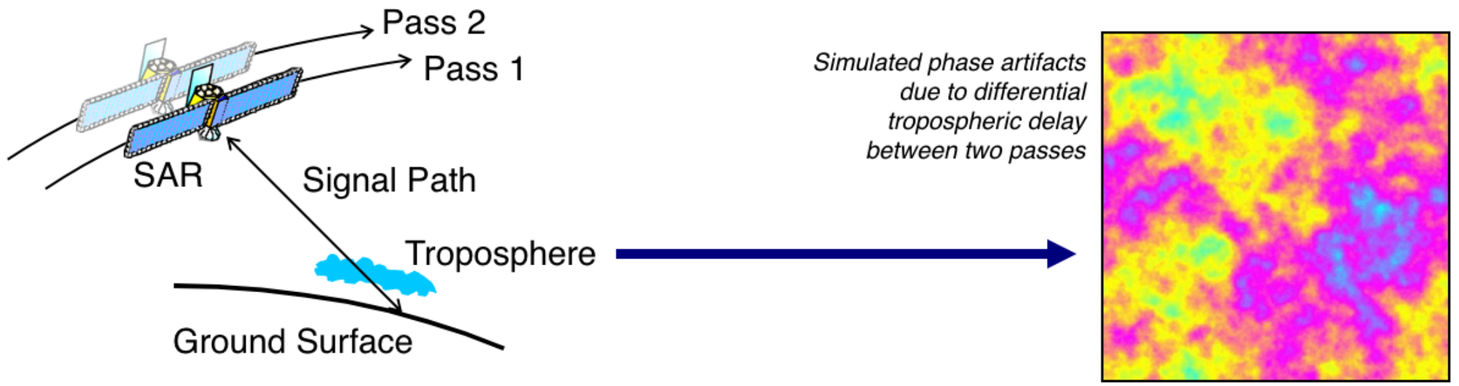
This was a major problem before 2000, when global topography was neither uniformly accurate nor available. In 2000, NASA's Shuttle Radar Topography Mission (SRTM) mapped topography between 60° S and 60° N.

Left: Etna processed with a pre-SRTM DEM; right: with SRTM. The red arrow marks a badly affected area, and many residual fringes, even near the centre of maximum displacement, come from the DEM.

ERROR SOURCES

Tropospheric noise

The troposphere’s refractive index varies in space and time, mainly because of water vapour. The radar wave picks up a spatially variable path delay, which becomes a phase delay, even with no ground displacement.



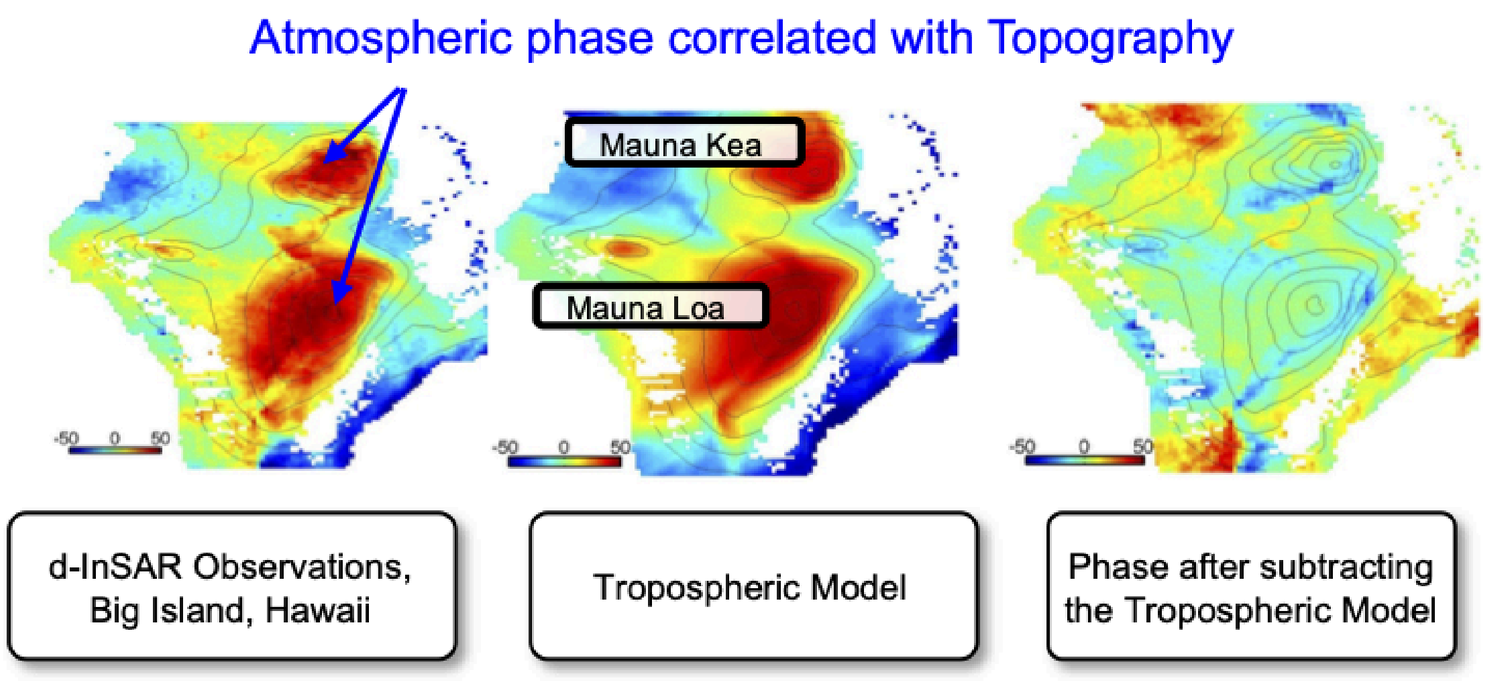
Simulated phase artifacts due to differential tropospheric delay between two passes

$$\Delta\phi(\rho, x) = \frac{4\pi}{\lambda}\left(\delta\rho_{\text{disp}}(\rho, x) + \delta\rho_{\text{atm}}(\rho, x)\right) + \text{noise}$$

Two unknowns that can’t be told apart. Mitigate with many interferograms spanning a common event: the atmosphere changes, persistent patterns are likely real.

ERROR SOURCES

Tropospheric noise example: Hawaii



Left: phase over the active volcanoes of the Big Island. Centre: phase predicted from the change in tropospheric delay between the two dates. Right: their difference. Most of the observed phase is tropospheric, not deformation.

HOMEWORK

Flat-Earth fringe rate

An interferometer flying over a flat Earth

Parameter	Symbol	Value
Wavelength	λ	0.24 m
Altitude	h	600 km
Baseline length	|b|	100 m
Baseline angle	α	−60°
Look angle, near range	θ_nr	30°
Look angle, far range	θ_fr	45°

Task

Calculate the ranges ρ_nr and ρ_fr, and the interferometric phase rate (fringe rate) ∂Δφ/∂ρ at each of these locations.

Use the fringe-rate formula from Part 2, with ψ_c = 0 for flat terrain:

$$\frac{\partial \Delta\phi}{\partial \rho} = \frac{2\pi p}{\lambda} \frac{b\cos(\theta - \alpha)}{\rho \tan(\theta - \psi_c)} = \frac{2\pi p}{\lambda} \frac{b_\perp}{\rho \tan(\theta - \psi_c)}$$

FURTHER READING

Other resources

01 Simons, M. & Rosen, P. (2015). Interferometric Synthetic Aperture Radar Geodesy. In Schubert, G. (ed.), *Treatise on Geophysics*, 2nd ed., Vol. 3: Geodesy, Elsevier, 339–385.

02 Rosen, P. A., Hensley, S., Joughin, I. R., Li, F. K., Madsen, S. N., Rodríguez, E. & Goldstein, R. M. (2000). Synthetic Aperture Radar Interferometry. *Proc. IEEE*, 88, 333–382.

03 Bürgmann, R., Rosen, P. & Fielding, E. (2000). Synthetic Aperture Radar Interferometry to measure Earth's surface topography and its deformation. *Ann. Rev. Earth Planet. Sci.*, 28, 169–209.

04 Madsen, S. N. & Zebker, H. A. (1999). Synthetic Aperture Radar Interferometry: Principles and Applications. *Manual of Remote Sensing*, Vol. 3, Artech House.

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COURSE MATERIALS ON GITHUB

Geo-SInC: the InSAR short-course repository

Repository

github.com/parosen/Geo-SInC

Free Jupyter notebooks from the EarthScope (formerly UNAVCO) InSAR Processing and Analysis short course, 2021–2024. Today's lecture is **1.3_InSAR_theory / InSAR.ipynb**.

PRE-COURSE

Python, GIS with GDAL, data access and SAR theory (Lecture 1)

DAY 1

Geophysical modelling, InSAR theory (this lecture), stripmap processing with ISCE2

DAY 2

Interpreting coseismic interferograms; TOPS (Sentinel-1) processing

DAY 3

Glaciers, preparing data for modelling, atmospheric errors

DAY 4

Time-series theory, stacks with ARIA-tools, offset stacks

DAY 5

Time-series analysis with MintPy and stack processing with ISCE2

Next in the same folder: Correlation.ipynb (interferometric decorrelation). To get everything, figures included: on GitHub, choose Code → Download ZIP.

Thank you. Questions?

Content adapted from Paul A. Rosen and Scott Hensley, “Geometric Aspects of InSAR”, InSAR short-course notebook (2021), github.com/parosen/Geo-SInC. Wave-interference figure © 2006 Encyclopædia Britannica, Inc.